An explicit polynomial in 29 variables over the rationals

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Abstract

We specify an integer polynomial in 29 total variables, with degree at most 2^{174} and coefficient height at most $2^{2^{177}}$. Under four precisely stated arithmetic inputs, its rational input projection is exactly $\mathbb{Z}$. A fixed prime forces a sufficiently large elliptic index through an existing scalar pole condition, so no separate size certificate is needed. Nine quartic point choices and four quadratic pole choices are eliminated by a finite-algebra construction using one additional rational coordinate. The paper gives the rational source, the local and real existence arguments, the integer-polynomial formula, and reproducible degree and coefficient bounds. The arithmetic premises are explicitly distinguished from the algebraic and computational conclusions proved here.

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1 The construction and its scope

For $F \in \mathbb{Z}[X, Y_1, \ldots, Y_m]$, its rational input projection is

$$\{x \in \mathbb{Q} : \exists \boldsymbol{y} \in \mathbb{Q}^m, F(x, \boldsymbol{y}) = 0\}.$$

We count the input among the variables. Degree means total degree; coefficient height means the largest absolute integer coefficient. The polynomial constructed below has $m = 28$ and satisfies

$$\deg F \le 2^{174}, \qquad H(F) \le 2^{2^{177}}.$$

These are upper bounds for a specified polynomial, rather than exact values of an expanded coefficient table. Its projection equals the integers under the four arithmetic inputs in Section 4. Those inputs are results asserted in the cited upstream manuscript and Sun's prime-pattern theorem. The implication from them is proved here; the coefficient checks do not establish the deep arithmetic inputs.

The exact specification comprises an integer coefficient expression graph and finite determinant, trace, resultant and product operations. Every range is explicitly bounded. The formal roots and matrix entries are internal to these finite operations and are not additional existential variables.

2 The exact numerical height constant

The contact-height input uses a fixed, explicitly specified integer. We record its exact size for the index estimates below. Put

$$B_* = \prod_{p \le 76176} p, \qquad H_{19} = 2^{110} 210^{1800} B_*^{19} + 128.$$

This is the constant of the source's exponent-nineteen height theorem [1, Theorem `thm:quartic-height19`].

Lemma 2.1. *The exact inequalities*

$$2^{2093447} < H_{19} < 2^{2093448}$$

hold. In particular one may use $h = 2093448$ in every argument requiring $H_{19} < 2^h$.

Proof. This finite integer calculation has the following reproducible certificate. Enumerate primes through 76176 by marking, for each unmarked $p \le \lfloor \sqrt{76176} \rfloor$, all multiples starting at p^2. The remaining integers at least two are exactly the primes: every composite has a prime divisor not exceeding its square root, and no prime is marked. An independent enumeration tests every integer by trial division through its integer square root. Both lists have 7499 entries and agree entry by entry.

Their exact product has 109445 binary digits. Computing the displayed powers and product as integers gives 2093448 binary digits for H_{19}. The comparison with the next smaller power of two is strict as well. The supplied checker performs both prime enumerations and both integer comparisons; the complete prime list and the unsigned binary encoding of H_{19} are included. No floating-point logarithm enters these comparisons. □

This lemma evaluates a constant. It does not prove the contact-height inequality itself. The latter is still the arithmetic input stated in Section 4, now used with this exact value and its sharper certified bound.

3 An explicit polynomial for a guarded finite algebra

Let $R = \mathbb{Z}[x_1, \dots, x_b]$. Consider a triangular monic presentation

$$A = R[z_1, \dots, z_m]/(f_1, \dots, f_m), \qquad f_i \in R[z_1, \dots, z_i],$$

where f_i is monic of degree $e_i \geq 1$ in z_i. Its standard monomials $\prod_i z_i^{a_i}$, $0 \leq a_i < e_i$, form an R-basis. In particular A is free of rank $n = \prod_i e_i$. For $a \in A$, write M_a for its multiplication matrix in this ordered monomial basis. The matrices M_a commute, and their entries are integer polynomials in the base coordinates. They are completely specified by repeated reduction using the monic equations; they need not be materialized to specify a determinant involving them.

Fix elements $G, E_1, \dots, E_r \in A$. We assume that in every geometric fiber, every point at which G is nonzero has local length one. This condition holds, for example, if G contains the product of the diagonal derivatives $\partial f_i/\partial z_i$ as a factor. We seek the rational base points admitting a rational point of A with

$$G \neq 0, \qquad E_1 = \cdots = E_r = 0.$$

For $0 \leq \ell \leq mn(n+1)/2$, define

$$\lambda_\ell = 1 + \sum_{i=1}^{m} \ell^i z_i, \qquad M = M_{G\lambda_\ell}, \qquad H(V) = \det(I - VM) = \sum_{k=0}^{n} a_k V^k.$$

Here and below $a_k \in R$ depend on the fixed integer ℓ; always $a_0 = 1$. Also put

$$R_i(V) = \operatorname{Tr}\big(M_{E_i} \operatorname{adj}(I - VM)\big).$$

This is a polynomial in $R[V]$ of degree at most $n - 1$.

For each $1 \leq d \leq n$, set

$$\begin{aligned}
H_d(V) &= \sum_{k=0}^{d} a_k V^k, \\
g &= a_d \operatorname{Res}_V(H_d, H_d'), \qquad L = a_d^2 g, \\
\Psi(U) &= 1 + \sum_{j=1}^{d} a_{d-j} a_d^{2j-1} g^j U^j, \\
\widetilde{R}_i(U) &= (LU)^{n-1} R_i(1/(LU)), \\
Q_{\ell,d}(\mathbf{x}, U) &= \Psi(U)^2 + \sum_{k=d+1}^{n} a_k^2 + \sum_{i=1}^{r} \widetilde{R}_i(U)^2.
\end{aligned}$$

The resultant is the Sylvester determinant with the specified degree bounds d and $d-1$. The expression for $\widetilde{R}_i$ denotes polynomial coefficient reversal, hence defines a polynomial also at $LU = 0$. Every displayed expression has integer coefficients.

Theorem 3.1. *The explicitly specified polynomial*

$$P(\mathbf{x}, U) = \prod_{\ell=0}^{mn(n+1)/2} \prod_{d=1}^{n} Q_{\ell,d}(\mathbf{x}, U)$$

has the following property. For every $\mathbf{a} \in \mathbb{Q}^b$, there exists $U \in \mathbb{Q}$ with $P(\mathbf{a}, U) = 0$ if and only if there is a rational point of the fiber algebra $A_{\mathbf{a}}$ satisfying $G \neq 0$ and all $E_i = 0$. The formula uses only one coordinate in addition to the b base coordinates; its formal algebra generators are eliminated by the determinants and are not quantified coordinates.

Proof. Fix a rational base point. First consider one factor $Q_{\ell,d}$. At $a_d g = 0$ or $U = 0$, its reciprocal residual Ψ equals one. Otherwise, with $V = 1/(LU)$,

$$\Psi(U) = (LU)^d \frac{H_d(V)}{a_d}.$$

Thus a rational zero of $Q_{\ell,d}$ implies $a_d g U \neq 0$, all coefficients a_k with $k > d$ vanish, and $H(V) = H_d(V) = 0$. The resultant condition makes this a simple root of H. Since $H(0) = 1$, the root V is nonzero.

The rational matrix $I - VM$ has a one-dimensional kernel. Every multiplication matrix M_{z_i} preserves it and therefore acts on it by a rational scalar ζ_i. Applying the algebra relations to this common eigenline shows that $(\zeta_1, \ldots, \zeta_m)$ is a rational point of the fiber. Its $G\lambda_\ell$-value is $1/V$, so its G-value is nonzero.

To verify the closed conditions, write $J = \mathrm{Tr}(\mathrm{adj}(I - VM))$. At a root of H, the adjugate identity gives $M \,\mathrm{adj}(I - VM) = V^{-1} \,\mathrm{adj}(I - VM)$. Differentiating the determinant then yields

$$J = -VH'(V) \neq 0.$$

The columns of the adjugate lie in the common eigenline, so

$$R_i(V) = E_i(\boldsymbol{\zeta}) J.$$

Since $LU \neq 0$, the equation $\widetilde{R}_i(U) = 0$ is equivalent to $R_i(V) = 0$, and hence $E_i(\boldsymbol{\zeta}) = 0$. This proves the reverse implication without assuming that the rest of the fiber algebra is reduced.

Conversely, suppose a suitable rational point exists. Over an algebraic closure, multiplication by $G\lambda_\ell$ has the value $G(p)\lambda_\ell(p)$ at every geometric point p, with multiplicity equal to its local length. If $G(p) = 0$, its factor in $\det(I - VM)$ is one, even if the local algebra is nonreduced. If $G(p) \neq 0$, its local length is one by hypothesis.

At any such point p, the polynomial $G(p)(1 + \sum_{i=1}^m \ell^i z_i(p))$ is nonzero as a polynomial in ℓ, because its constant coefficient is $G(p) \neq 0$. For distinct such points p, q, their difference is also nonzero: if all coefficients vanished, first $G(p) = G(q) \neq 0$, and then all their coordinates would agree. Each of these polynomials has degree at most m. With at most n geometric points, there are at most $mn(n+1)/2$ bad values of ℓ in total, counting zero values and collisions. Thus one of the specified integers makes all nonzero guarded eigenvalues distinct and nonzero.

For that ℓ, the polynomial H is squarefree of some degree $1 \leq d \leq n$. The given rational point provides a nonzero rational root V of H. Its closed residuals vanish, so the trace identities give $R_i(V) = 0$. The leading coefficient and resultant are nonzero. Taking $U = 1/(LV)$ therefore gives a zero of the corresponding factor $Q_{\ell,d}$, and hence of P. □

The degree cases are part of the literal finite formula. They include every exceptional drop in degree of H. The discriminant condition excludes bad separating choices, and the finite bound on ℓ proves that this exclusion loses no valid rational point. In particular the construction does not leave a generic primitive-element argument or an unperformed exceptional-stratum decomposition implicit.

4 Odd products and a 29-variable construction

Variable counts include the distinguished input. The construction uses Conant's odd-product height identity and direct-target comparison argument as presented in [1, Lemma `v9:lem:product` and Lemma `lem:v9-bridge`], together with the explicit finite algebra construction. We give the rational system, prove its input projection under precisely stated arithmetic hypotheses, and then eliminate its finite algebraic choices. The product identity and point charts are proved independently of the contact-height and prime-pattern inputs.

4.1 Arithmetic inputs and conventions

Put

$$K = \mathbb{Q}(\sqrt{2}), \quad d = 3 - \sqrt{2}, \quad D = d^2 = 11 - 6\sqrt{2}, \qquad E : y^2 = x^3 - Dx,$$

and fix

$$R_0 = (1 + \sqrt{2}, 2\sqrt{2}), \qquad P = 2R_0 = \left(\frac{57}{8} - \frac{7}{2}\sqrt{2}, -\frac{165}{8} + \frac{417}{32}\sqrt{2}\right).$$

Write σ for conjugation, $T_0 = (0,0)$, $f(Q) = 1/x(Q)$, and $z(Q) = -x(Q)/y(Q)$. Heights are absolute logarithmic Weil heights, with normalized local degree weights. Valuations at unramified primes satisfy $v(p) = 1$.

We use the following arithmetic hypotheses, with the named source locations in [1].

1. (Elliptic group and height.)
   Source: `subsec:v5-point-source`, `eq:o8-indexheight`.

   $2E(K) = \mathbb{Z}P$, and $h(x([n]P)) \geq n^2 - 6$ for every nonzero integer n.

2. (Contact height: Theorem `thm:quartic-height19`.) For $t = u/v$ in lowest terms, $v > 0$, and $t \notin \mathcal{D} = \{0, 1, 2401, -343/32\}$, define $M_5(t)$ to be the product of the primes $p > 76176$ for which at least one of

   $$v_p(v), \quad v_p(u), \quad v_p(u - v), \quad v_p(u - 2401v), \quad v_p(32u + 343v)$$

   is positive and odd. These are the normalized contacts with $\infty, 0, 1, 2401, -343/32$. Put

   $$B_* = \prod_{p \leq 76176} p, \qquad H = H_{19} = 2^{110} 210^{1800} B_*^{19} + 128.$$

   The inherited assertion is $h(t) \leq H M_5(t)^{19}$. Exact integer evaluation gives $H < 2^{2093448}$. We use this exponent without weakening it, so $\eta = 19$.

3. (Pole obstruction and realization.)
   Source: `hyp:o8-pole` and `lem:original-seven-simultaneous-realization`, and `v5:preparation-zero-bridge`. The pole equations, with $B = 17b$,

   $$U = b\frac{1 - bt^2}{1 + bt^2}, \quad h_0 = e(B - U^2), \quad p^2 = B(1-e)(1+3e), \quad Bq^2 = (s^2 - Bh_0)(s^2 + 3Bh_0),$$

   exclude negative odd $v_\ell(b)$ for $\ell \notin \{2, 3, 17\}$. They are soluble with $t = 0$ whenever b has square reduced denominator, is positive and less than 17, is square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, and satisfies $v_2(b) \geq 3$, $v_3(b) \geq 1$, $v_{17}(b) \geq 2$.

4. (Simultaneous prime patterns: Theorem `thm:v5-simultaneous-primes`, Lemma `lem:v5-two-reductions`, and the application of [3, Theorem 4.1].) The simultaneous Gaussian prime-pattern input supplies the prescribed prime values $u + c_j w^2$, arbitrary fixed divisibility of w, and arbitrarily large u/w^2. Their two conjugate reduction orders satisfy

   $$v_2(d_-) = 6, \quad v_2(d_+) = 2, \quad \gcd(d_-(p), d_+(p')) = 4$$

   for every pair of selected primes. An arbitrary fixed odd avoidance modulus can be imposed. For each selected Gaussian prime π, the source convention is $\pi \equiv 1 + 3i \pmod 7$, making both $N(\pi - 1)$ and $N(\pi + 1)$ units at seven. At each other odd prime ℓ in the fixed avoidance modulus, the condition $\pi \equiv i \pmod \ell$ makes both orders congruent to two modulo ℓ. Thus the orders avoid the entire odd part of the fixed modulus. The selected rational primes are 1 modulo $8 \cdot 3 \cdot 17$.

These hypotheses include the simultaneous application of Sun's stated prime-pattern theorem. They are assumptions of the theorem below, not conclusions of its implementation audit. The corrected arithmetic proof of the pole realization is the one in Version 11 or later of the source manuscript [2]. The finite-avoidance and zero-preparation consequences are identified precisely in the accompanying arithmetic dependency paper.

The assigned comparison orders are $\mu = 10$ and $\kappa = 39$. The logarithm truncation parameters are $\rho_0 = 12$ and $\rho_1 = 40$, and the scalar ratio order is $m = 20$. The fixed-prime argument in Section 5 supplies the needed index size.

4.2 An elementary Lucas growth estimate

The following estimate is independent of the arithmetic hypotheses. It supplies the growth inequality used in the integer-extraction argument.

Lemma 4.1. *Let $a, t, b \in \mathbb{Z}$, with $|a| > 7$ and $b \neq 0$, satisfy*

$$t^2 - a(a-2)tb + (a-2)^2b^2 = 1.$$

Then $\log|t| > |a| - 3$. In particular, $\log\max(|a|, |t|, |b|) > |a| - 3$.

Proof. Put $A = |a|$, $c = (a-2)b$ and $c' = \operatorname{sgn}(a)c$. Then

$$t^2 - Atc' + (c')^2 = 1.$$

We have $|c'| \geq |a-2| \geq 6$, so $t \neq 0$. The two terms t, c' have the same sign: opposite signs would make the left-hand side at least $t^2 + (c')^2 > 1$. Thus $T = |t|$ and $C = |c'|$ are positive integer solutions of $T^2 - ATC + C^2 = 1$.

Define $U_0 = 0$, $U_1 = 1$ and $U_{j+1} = AU_j - U_{j-1}$. Every positive solution consists of adjacent terms of this sequence. Indeed, order its coordinates as $M \geq m > 0$. The integer

$$Am - M = \frac{m^2 - 1}{M}$$

lies in $[0, m)$, and replacing (M, m) by $(m, Am - M)$ preserves the equation. Repeated descent ends at $(1, 0)$. Reversing the steps gives the unordered pairs $\{U_{j+1}, U_j\}$. Consequently $C = U_j$ for some $j \geq 1$ and $T \geq U_{j-1}$.

Put $d = |a-2|$. Since $d \mid C$, reduction of the recurrence modulo d gives $d \mid j$: if $a > 0$, then $A \equiv 2 \pmod{d}$ and $U_j \equiv j \pmod{d}$; if $a < 0$, then $A \equiv -2 \pmod{d}$ and $U_j \equiv (-1)^{j-1}j \pmod{d}$. Hence $j \geq d \geq A - 2$. The recurrence also gives $U_r > (A-1)^{r-1}$ for $r \geq 2$. This follows by induction from $U_2 = A > A - 1$ and $U_{r+1} > (A-1)U_r$, since $U_r > U_{r-1}$. As $j \geq 6$, we obtain

$$\log T > (j-2)\log(A-1) \geq (A-4)\log 7 > A - 3 \qquad (A \geq 8).$$

The last inequality holds at $A = 8$ and its difference increases with A. This proves the assertion. □

4.3 An exact height identity

Lemma 4.2. *For any nonzero odd integers $n_1, \ldots, n_r$,*

$$h\left(\prod_{i=1}^{r} x([n_i]P)\right) = \sum_{i=1}^{r} h(x([n_i]P)).$$

This identity requires only that the displayed $P = 2R_0$ have infinite order; it does not require the contact-height or prime-pattern hypotheses.

Proof. First consider a finite place v with $v(D) = 0$. The formal subgroup $E_1(K_v)$ consists of the identity and points with $v(x) < 0$. It is an open subgroup, so $E(K_v)/E_1(K_v)$ is finite. These formal-group facts hold for the displayed integral Weierstrass equation, including the place above two [4, Chapter VII]. The image of T_0 has order two, since $T_0 \notin E_1(K_v)$, and

$$x(Q + T_0) = -D/x(Q).$$

If an odd multiple of P belongs to $E_1(K_v)$, the image of P in the quotient has odd order. No multiple of that image is the nonzero element represented by T_0. Hence no odd multiple of P has positive x-valuation. If no odd multiple belongs to $E_1(K_v)$, all odd multiples have nonnegative x-valuation. Thus the numbers $v(x([n_i]P))$ have a common weak sign.

The only remaining finite place is the prime $(3 - \sqrt{2})$ above seven. Here $v(D) = 2$, and the discriminant has valuation six, so the displayed integral model is minimal. Its reduction is $y^2 = x^3$. The displayed P reduces to $(1, 6)$, a nonsingular point. The subgroup of points with nonsingular reduction therefore contains every multiple of P. A point with positive x-valuation would have positive y-valuation and reduce to the singular point $(0, 0)$. Consequently every multiple of P has x-valuation at most zero at this place.

At either real embedding every $[n_i]P$ is a nonzero double. The duplication identity

$$x(2Q) = \left(\frac{x(Q)^2 + D}{2y(Q)} \right)^2$$

makes its abscissa positive. The curve equation and its nonzero ordinate imply $x([n_i]P) > d > 1$ in each embedding. Each real local height therefore adds under multiplication. At a finite place the common weak sign similarly gives

$$\max\{0, -\textstyle\sum_i v(x([n_i]P))\} = \sum_i \max\{0, -v(x([n_i]P))\}.$$

Summing with the normalized local weights proves the identity. □

Corollary 4.3. *Let $Q_1, \dots, Q_4$ be odd multiples of P and write*

$$C = \prod_{i=1}^{4} x(Q_i) = C_0 + C_1\sqrt{2}.$$

The three numbers $A_0 = 1$, $A_1 = C_0^2$, and $A_2 = (1 + C_1^2)^2$ are nonzero positive rational squares. If the three contact detectors at $A_j\lambda$ imply $M_0(A_j\lambda)^k \mid N \neq 0$, then

$$h(x(Q_i)) < 7H|N|^{\eta/k} \qquad (1 \le i \le 4).$$

Here $\Theta = \{0, -1/2, -1, 1/2399, -32/407\}$, and $M_0(t)$ is the product of the primes $p > 76176$ for which $v_p(t - \theta)$ is positive and odd for at least one $\theta \in \Theta$.

Proof. The product is positive in both embeddings, so $C_0 > 0$. The fractional linear map $t \mapsto 2 + 1/t$ identifies the five contacts with the original five, outside the exceptional primes. Consequently $h(t) \le (H + \log 3)M_0(t)^\eta$. Put $T = (H + \log 3)|N|^{\eta/k}$. The unscaled detector and each scaled detector give $h(A_j) \le 2T$. Hence $h(C_0) \le T$. For $C_1 = u/v$ in lowest terms with $v > 0$, $h(1 + C_1^2) = \log(u^2 + v^2) \ge 2h(C_1)$, so $h(C_1) \le T/2$. The elementary height inequality in the fixed basis gives

$$h(C) \le \tfrac{3}{2}T + \tfrac{3}{2}\log 2 < 7H|N|^{\eta/k}.$$

Lemma 4.2 bounds every summand by $h(C)$. □

4.4 Point charts with no free parity witnesses

Lemma 4.4. *Suppose $R = (a, b) \in E(K)$ is not a two-torsion point. At least one of $R, R + T_0$ has nonzero rational component of its ordinate. Thus every nonzero double is represented by $2R$ in the chart below, and every odd multiple $[n]P$ with $n \neq 1$ is represented by $P + 4R$ in the same chart.*

Proof. If both ordinates had zero rational component, then $b \in \sqrt{2}\mathbb{Q}^\times$ and $D/a^2 = b(R + T_0)/b(R) \in \mathbb{Q}$. Therefore $a/d \in \mathbb{Q} \cup \sqrt{2}\mathbb{Q}$. Since $b^2 = d^3(a/d)((a/d)^2 - 1) \in \mathbb{Q}^\times$, this would force $d^3 \in \mathbb{Q} \cup \sqrt{2}\mathbb{Q}$, contrary to $d^3 = 45 - 29\sqrt{2}$. Translation by T_0 preserves both $2R$ and $4R$.

If n is odd and $n \neq 1$, take $R = [(n - 1)/2]R_0$, so $P + 4R = [n]P$. Its $4R$ is nonzero. The affine addition denominator $x(4R) - x(P)$ cannot vanish: otherwise $4R = \pm P$, whereas $4E(K) = 2\mathbb{Z}P$ contains only even multiples. The same translation by T_0 ensures the chosen chart is defined. □

For each of the nine names $A, S, T, Z, H_0, U_A, U_S, U_T, U_Z$, retain a rational pair (a_0, a_1). Write $a = a_0 + a_1\sqrt{2}$ and $a^3 - Da = A' + B'\sqrt{2}$. Introduce a finite algebra root Y with

$$Y^4 - 4A'Y^2 + 8(B')^2 = 0, \qquad b = Y/2 + (B'/Y)\sqrt{2}.$$

Require $Y \neq 0$, $(A')^2 - 2(B')^2 \neq 0$, and $4Y^3 - 8A'Y \neq 0$. Then $R = (a, b) \in E(K)$. For the four root names set the output point to $P + 4R$; for the other five set it to $2R$. All arithmetic is by the usual affine addition and duplication formulas with their denominators guarded. The derivative condition is automatic at rational points of this chart but is retained to ensure geometric simplicity in the finite algebra. Indeed, since $Y \neq 0$, a vanishing derivative would give $Y^2 = 2A'$; the quartic would then give $(A')^2 = 2(B')^2$, forcing $A' = B' = 0$ over $\mathbb{Q}$, contrary to the norm guard. It introduces no extra coordinate.

In particular, put $C_S = S - P = 4R_S$. This point, rather than S, will be used in the scalar pole certificate. Write

$$(A, S, T, Z) = ([n]P, [a]P, [t]P, [b]P).$$

The four labels are nonzero odd integers; the labels are consequences of the group hypothesis, not additional quantified variables.

4.5 The rational system

Retain the input X, the eighteen point-chart coordinates, one rational λ, two parameters for each of three comparison poles, and two parameters for a scalar pole. There are

$$1 + 18 + 1 + 6 + 2 = 28$$

retained rational coordinates. Set

$$\beta = 216X^4 + 450459.$$

The following elementary input map eliminates every scalar correction choice.

Lemma 4.5. *For rational X, the number $\beta = 216X^4 + 450459$ is integral if and only if X is integral. It is greater than 2401. For integral X, its contact $r = \beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1)$ is a square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, with $v_2(r) = v_3(r) = 2$ and $v_{17}(r) \in \{0, 2\}$.*

Proof. This is proved, with its complete residue table and exceptional valuation calculation, in Lemmas 6.1 and 6.2. □

Put $Q_0 = P$, $(Q_1, Q_2, Q_3, Q_4) = (A, S, T, Z)$, $U_0 = H_0$, and $V_i = U_i - Q_i$ for $0 \le i \le 4$. For $k \in \{\rho_0, \rho_1\}$ define

$$\ell_k(Q) = z(Q) \sum_{j=0}^{k/4-1} c_j (Df(Q)^2)^j, \quad c_0 = 1, \quad c_1 = -2/5, \quad c_j = c_{j-1} \frac{4j-5}{4j+1}.$$

Put $(k_1, k_2, k_3, k_4) = (\mu, \kappa, \kappa, \kappa)$ and $(r_1, r_2, r_3, r_4) = (\rho_0, \rho_1, \rho_1, \rho_1)$. Set

$$L_i = \frac{\ell_{r_i}(U_i)}{\ell_{\rho_1}(H_0)}, \qquad L_i' = \sigma\left(\frac{\ell_{r_i}(V_i)}{\ell_{\rho_1}(V_0)}\right).$$

The sixteen field-valued tests, with their orders, are

$f(U_0), \dots, f(U_4), \sigma f(V_0), \dots, \sigma f(V_4)$	1 each
$L_i - L_i'$	k_i
$L_1 - \beta$	μ
$L_3^2 - L_2(L_2-2)L_3L_4 + (L_2-2)^2 L_4^2 - 1$	κ.

Denote these tests by D_j and their orders by m_j. Use precisely the three scales of Corollary 4.3. For each such A_r put

$$t_r = A_r\lambda, \quad \delta_r = t_r - 1, \quad F_r = \prod_{\theta \in \Theta} (t_r - \theta), \quad R_r = \frac{F_r}{A_r^5 \delta_r^6},$$

$$W_r = \prod_{j=1}^{16} \left(1 + R_r \left(\frac{D_j}{\delta_r^{m_j+1} R_r^{m_j}}\right)^2\right), \qquad b_r = \frac{R_r}{\mathrm{Tr}_{K/\mathbb{Q}}(W_r^{-1})^2}.$$

All F_r, δ_r, norms of W_r, and displayed denominators are required to be nonzero. In the coefficient DAG, if $W_r = w_0 + w_1\sqrt{2}$, the last expression is exactly

$$b_r = \frac{F_r(w_0^2 - 2w_1^2)^2}{4A_r^5 \delta_r^6 w_0^2}.$$

For each nonzero pole input b use two retained parameters τ, s and a fixed finite choice $\varepsilon \in \{0, 1\}$. Set

$$B = 17b, \quad e = \frac{\tau^2 + (4\varepsilon - 1)B}{\tau^2 + 3B}, \quad h_0 = e(B - b^2), \quad q^2 = \frac{(s^2 - Bh_0)(s^2 + 3Bh_0)}{B}.$$

Require $\tau^2 + 3B \neq 0$. The first original conic is recovered by $p = -4B\tau(1-\varepsilon)/(\tau^2 + 3B)$. Conversely every point of that conic is covered: if $e \neq 1$ take $\tau = p/(e-1)$ and $\varepsilon = 0$; if $e = 1$ take $\varepsilon = 1, \tau = 0$. This includes $e = -1/3, p = 0$. There is no nonzero condition on τ, p, or q.

For the fourth, scalar pole put

$$r_\beta = \beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1).$$

$$\delta_\beta = 1 + (17\ell^k r_\beta)^2, \quad R_\beta = \frac{102^2 \ell^{2k+1} r_\beta}{\delta_\beta^2}.$$

Here $\ell = 76249$ and $k = 137715$; the same constants are fixed in every occurrence of the scalar contact. Require $R_\beta \neq 0$. Write the rational components of $f(H_0), f(C_S), z(C_S)/z(H_0)$ as $g_0, \dots, g_5$, in that order. For $N_R(u, v) = u^2 + Rv^2$, set

$$w_0 = N_R(1, g_0/R), \quad w_1 = N_R(g_1/R, g_2/R), \quad w_2 = g_3/R, \quad w_3 = N_R(g_4/R^m, g_5/R^m),$$

$$W = N_R(N_R(w_0, w_1), N_R(w_2, w_3)), \qquad R = R_\beta.$$

Require $W \neq 0$, and give $R_\beta W^2$ the same pole certificate. There is no scalar correction parameter. The only original finite choices are the four conic bits, hence 2^4 source branches. For rational input $R_\beta > 0$, the constant-left norm tree in fact has $W \ge 1$, so this nonzero condition is automatic; it is harmless and convenient to retain the same denominator-guard convention.

4.6 Local implications and soundness

We include the local argument to make the effect of direct targets explicit. At a prime $p > 76176$ where $v_p(R)$ is odd, consider $W = \prod_j (1 + R(D_j/R^{m_j})^2)$, $T = \mathrm{Tr}(W^{-1})$, and a pole certificate for R/T^2. Each factor has valuation

$$\min\{0, v_p(R) + 2(v_{\mathfrak{p}}(D_j) - m_j v_p(R))\},$$

since its two terms have valuations of different parity. If $\lambda = \max_{\mathfrak{p}|p} v_{\mathfrak{p}}(W) \le 0$, then $v_p(T) \ge -\lambda$. The pole implication gives

$$0 < v_p(R/T^2) \le v_p(R) + 2\lambda.$$

Thus $v_p(R) > 0$ and $\lambda > -v_p(R)/2$. At a place attaining the maximum, every factor valuation is at least λ, whence $v_{\mathfrak{p}}(D_j) > (m_j - 3/4)v_p(R) \ge m_j - 3/4$ and therefore $v_{\mathfrak{p}}(D_j) \ge m_j$. For the normalized detectors, δ_r is a unit at each consumed contact prime, so this argument applies to the original D_j.

Write $H_0 = [h]P$ and $U_i = [u_i]P$. The first ten tests place the two families in the corresponding formal groups. Formal logarithms have exact ratios u_i/h and $(u_i - n_i)/(h-1)$. For the two truncation lengths $J = 3, 10$, the relative errors have valuation at least $4J$. Here is a direct denominator check. The exact formula $c_j = 2\prod_{i=1}^{j}(4i-5)/(4i+1)$ gives $v_p(c_j) \ge -\lfloor \log_p(4j+1) \rfloor \ge -j$ for every odd p: the two progressions have counts differing by at most one at each prime power. Terms with $j \ge \lceil 4J/3 \rceil$ therefore have relative valuation at least $3j \ge 4J$. For the remaining omitted terms $J \le j < \lceil 4J/3 \rceil$, every denominator factor is less than $p > 76176$, so their valuations are at least $4j \ge 4J$. Thus both ratios have the required precision k_i. At least one of $h, h-1$ is a p-unit. The associated ratio is integral; the cross-test of order k_i makes both ratios integral. Relative precision is therefore absolute precision. Multiplication by h and $h-1$ and subtraction gives

$$L_i \equiv n_i \pmod{p^{k_i}}.$$

This argument also holds at an inert prime. Integer polynomial congruences then give the corresponding congruence for the direct Lucas residual.

The scalar certificate gives

$$M_5(\beta)^m \mid (a-1), \qquad h(\beta) \le H|a-1|^{\eta/m}.$$

Indeed negative valuations of $\beta = 216X^4 + 450459$ are multiples of four outside the fixed primes two and three. At every consumed finite odd contact, R_β has positive odd valuation ν; its normalizer and fixed multiplier are units. The prime ℓ is never a contact counted by $M_5(\beta)$, as proved below. For N_R there is no cancellation between the two terms. If a leaf at depth j contributes to the root, its valuation has the form $2^j v(\mathrm{leaf}) + e\nu$, $0 \le e < 2^j$. The pole implication forces the root valuation to exceed $-\nu/2$; hence every normalized leaf has valuation greater than $-\nu$. An original test assigned order m consequently has valuation at least $(m-1)\nu + 1 \ge m$. The order-one tests put H_0, C_S in the formal group; the order-m ratio implies $v_p((a-1)/h) \ge m$, and thus $p^m \mid a-1$. The contact-height hypothesis proves the stated height bound.

Set $c = |a-1|$, $B = \max(|n|, |a|, |t|, |b|)$ and $\Delta = t^2 - a(a-2)tb + (a-2)^2 b^2 - 1$. If $\Delta \ne 0$, the local implications and the three product-height scales give

$$B^2 - 6 < 7H(5B^4)^{\eta/\kappa}.$$

The index size forced in Section 5 contradicts this inequality. Hence $\Delta = 0$ and $\log B > |a| - 3 \ge c - 4$. If $\beta \ne n$, let N be the absolute value of the numerator of $\beta - n$ in lowest terms. It satisfies $N \le 2e^{h(\beta)}B$, and each contact radical has its μth power dividing N. Therefore

$$B^2 - 6 < 7H(2e^{h(\beta)}B)^{\eta/\mu}.$$

The scalar estimate and the inequalities in that section again contradict Lucas growth. Thus $\beta = n \in \mathbb{Z}$, and Lemma 4.5 implies $X \in \mathbb{Z}$.

5 A fixed-prime contact and a construction with 29 variables

The size of the scalar elliptic index can be forced by the scalar pole condition itself. This removes the separate size certificate. We give all constants and the local argument explicitly.

Throughout this section let

$$K = \mathbb{Q}(\sqrt{2}), \qquad D = 11 - 6\sqrt{2}, \qquad E : y^2 = x^3 - Dx,$$

and let P be the normalized elliptic source point. Put

$$H = 2^{110}210^{1800}\left(\prod_{p\le 76176} p\right)^{19} + 128, \qquad H < 2^h, \qquad h = 2093448.$$

The four arithmetic inputs are the cited elliptic-group and height results, the seven-equation pole result, and Sun's simultaneous prime-pattern theorem. The use of those inputs, including arbitrary fixed finite avoidance and zero preparation, is established in the arithmetic dependency argument. No additional global existence assertion is introduced below.

5.1 An explicit prime with no contact root

For the rational input X define

$$\beta = 216X^4 + 450459, \qquad r = \beta(\beta-1)(\beta-2401)(32\beta+343)(2\beta-1).$$

In particular $\beta > 2401$ and $r > 0$ for every rational X. Fix

$$\ell = 76249, \qquad k = 137715.$$

For $\theta = 0, 1, 2401, -343/32, 1/2$, respectively, the five elements $(\theta - 450459)/216$ of $\mathbb{F}_\ell$ are

$$32862,\ 32509,\ 24048,\ 904,\ 70810.$$

Their powers with exponent $(\ell-1)/4$ are respectively

$$76248,\ 76248,\ 76248,\ 76248,\ 40177.$$

None is 1 or 0. The integer ℓ is prime: division by every integer from 2 through $276 = \lfloor\sqrt{\ell}\rfloor$ gives a nonzero remainder. These finite calculations, including the complete trial-division test, are reproduced by the supplied independent integer checker. They imply that none of the five equations $\beta = \theta$ has a solution in $\mathbb{F}_\ell$. Notice that the assertion needed here is failure to be a fourth power, not failure to be a square.

Lemma 5.1. *For every $X \in \mathbb{Q}$,*

$$v_\ell(r) = \begin{cases} 0, & v_\ell(X) \ge 0, \\ 20v_\ell(X), & v_\ell(X) < 0. \end{cases}$$

Define

$$R = \frac{102^2\ell^{2k+1}r}{\left(1 + (17\ell^k r)^2\right)^2}. \tag{1}$$

Then $\nu = v_\ell(R)$ is odd and satisfies

$$\nu \ge k + 16 = 137731.$$

Proof. If X is ℓ-integral, the preceding five fourth-power tests show that every factor of r is an ℓ-adic unit. If $v_\ell(X) < 0$, the term $216X^4$ strictly dominates the constant term in every factor. The coefficients $216, 2, 32$ are ℓ-adic units, so each factor has valuation $4v_\ell(X)$.

Write $t = v_\ell(r) \in 20\mathbb{Z}_{\le 0}$. Because $k \equiv 15 \pmod{20}$, the number $k + t$ is nonzero. Consequently the two summands of $1 + (17\ell^k r)^2$ have different valuations, and

$$\nu = 2k + 1 + t - 4\min(0, k + t).$$

This is odd. Write $k = 20q + 15$ and $t = -20j$, $j \ge 0$. For $j \le q$ the expression is $2k + 1 - 20j$, decreasing to $k + 16$. For $j \ge q + 1$ it is $1 - 2k + 60j$, increasing from $k + 16$. This proves the stated minimum for every rational input, including inputs with an ℓ-adic denominator. □

5.2 The scalar norm forces the necessary index size

Set $C_S = S - P = [s]P$ and $H_0 = [u]P$. Both points are nonzero in the source, so s, u are nonzero integers. As usual write $f = 1/x$ and $z = -x/y$. Let $g_0, \dots, g_5$ be, in order, the rational components of $f(H_0)$, $f(C_S)$ and $z(C_S)/z(H_0)$ in the basis $1, \sqrt{2}$. For $N_R(v, w) = v^2 + Rw^2$ set

$$\begin{aligned} w_0 &= N_R(1, g_0/R), & w_1 &= N_R(g_1/R, g_2/R), \\ w_2 &= g_3/R, & w_3 &= N_R(g_4/R^{20}, g_5/R^{20}), \end{aligned}$$

$$W = N_R\big(N_R(w_0, w_1), N_R(w_2, w_3)\big).$$

The scalar pole input is RW^2, with preparation parameter zero.

Lemma 5.2. *The scalar pole condition implies*

$$v_\ell(s) \ge 2616890, \qquad |s| > (64H)^{20}.$$

Proof. Because $\nu = v_\ell(R)$ is odd, the terms v^2 and Rw^2 have different valuations whenever both are nonzero. Thus a norm node has valuation equal to the smaller of the two term valuations. As ℓ is larger than 76176, the pole obstruction applied to RW^2 gives

$$v_\ell(W) > -\nu/2.$$

Every leaf contribution at depth j has valuation $2^j v_\ell(\text{leaf}) + e\nu$ with $0 \le e < 2^j$. It follows that every normalized leaf has valuation greater than $-\nu$. In particular

$$v_\ell(g_4),\ v_\ell(g_5) > 19\nu.$$

The order-one leaves similarly have positive valuation and put both H_0 and C_S in the formal group at every place above ℓ. The formal logarithm divided by z is a unit at such a point. Its exact scalar ratio is s/u, so

$$v_\mathfrak{l}\big(z(C_S)/z(H_0)\big) = v_\ell(s/u).$$

The components of this ratio both have valuation greater than 19ν, hence $v_\ell(s/u) \ge 19\nu + 1$. Since $u \in \mathbb{Z}$,

$$v_\ell(s) \ge 19\nu + 1 \ge 19 \cdot 137731 + 1 = 2616890.$$

Finally $\ell > 2^{16}$ and

$$16 \cdot 2616890 = 41870240 > 41869080 = 20(h + 6).$$

Therefore $|s| \ge \ell^{2616890} > 2^{20(h+6)} > (64H)^{20}$. □

For clarity, the numerical implication from this index size to soundness can be checked directly. Use assigned comparison orders $\mu = 10$ and $\kappa = 39$, scalar order $m = 20$, and formal-logarithm precisions 12 and 40. Put $c = |s| = |a - 1|$ and $B = \max(|n|, |a|, |t|, |b|)$. The local detector and height arguments give

$$\begin{aligned} h(\beta) &\le Hc^{19/20}, \\ \Delta \neq 0 &\implies B^2 - 6 < 7H(5B^4)^{19/39}, \\ \beta \neq n &\implies B^2 - 6 < 7H(2e^{h(\beta)}B)^{19/10}, \end{aligned}$$

where $\Delta = t^2 - a(a-2)tb + (a-2)^2b^2 - 1$. The first inequality and $c > (64H)^{20}$ imply $h(\beta) < c/64$. If $\Delta \neq 0$, then $5^{19/39} < \sqrt{5} < 16/7$ and $B > \sqrt{12}$ give $B^{2/39} < 32H$. But

$$B \ge |a| \ge c - 1 > c/2 > (64H)^{20}/2 > (32H)^{39/2},$$

a contradiction. Hence $\Delta = 0$, and the proved elementary Lucas growth lemma gives $\log B > c-4$. If $\beta \neq n$, the last displayed implication gives

$$\log B < 10 \log(64H) + 19h(\beta) < 29c/64 < c - 4,$$

again a contradiction. In the second inequality we used $\log(64H) < c/64$, which follows from $H \ge 1$ and $c > (64H)^{20}$. Thus $\beta = n \in \mathbb{Z}$ and the input-map integrality lemma gives $X \in \mathbb{Z}$.

The scalar height argument used here still applies at every prime counted by $M_5(\beta)$. Indeed, ℓ is never such a prime: at ℓ the input either has no contact or its denominator valuation is a multiple of four. At any other consumed contact prime, the extra factor ℓ^{2k+1} is a unit and the new normalizer is a unit. The valuation of R is therefore the same positive odd valuation as for r. The norm-tree argument gives $p^{20} \mid s$, as required.

5.3 Realization of the scalar pole for every integer input

Lemma 5.3. *For every integer X, the scalar normalization in (1) permits the simultaneous source construction with zero preparation. It adds no retained variable, finite generator, or independent arithmetic hypothesis.*

Proof. For integer X the input-map congruences give a positive integer r which is a square in $\mathbb{Q}_2$, $\mathbb{Q}_3$ and $\mathbb{Q}_{17}$, with

$$v_2(r) = v_3(r) = 2, \qquad v_{17}(r) \in \{0, 2\}.$$

Since $\ell \equiv 1 \pmod{24}$ and $\ell \equiv 4 \pmod{17}$, the integer ℓ is itself a square in each of these three local fields. The normalizer $\delta = 1 + (17\ell^k r)^2$ is an integer and a unit at $2, 3, 17$. Moreover $\gcd(r, \delta) = 1$ and $\delta \equiv 1 \pmod{\ell}$, so its denominator has no common prime factor with the numerator of R. Thus R has square reduced denominator, has the same three local squareclasses as r, and has valuations at least $4, 4, 2$ respectively. Also

$$0 < R < \frac{102^2}{17^4} \frac{\ell^{1-2k}}{r^3} < 1.$$

The first inequality uses $(1 + y^2)^2 > y^4$; the last uses $102^2 < 17^4$, $r \ge 1$ and $k \ge 1$.

Fix this integer input before choosing the elliptic indices. The set of prime divisors of the numerator or denominator of R is finite; it includes ℓ. Choose H_0 in a small compact neighborhood of the real identity with $z(H_0)$ nonzero, and sufficiently deep in the formal groups at this finite set. Choose a positive s, divisible by the prescribed fixed modulus and by six, still deeper at these places, and with $[s]P$ sufficiently close to the real identity. The density of each prescribed compatible progression in the identity component of the compact real closure supplies these choices. The six normalized leaves can then be made arbitrarily small at the real

places and arbitrarily deep at the relevant finite places. Consequently W is a unit at every prime in the support of R and is sufficiently close to 1 in the real absolute value. In particular one may arrange $|W-1|<1/2$, so $0<RW^2<9/4<17$.

The scalar pole input RW^2 has square reduced denominator: outside the fixed support its valuation is twice that of W, and on the support W is a unit. At $2,3,17$ it retains the required local squareclasses and depths. The zero-preparation pole theorem therefore supplies its rational certificate. Its first conic is covered by the two specified rational charts, including its exceptional point.

The remaining simultaneous construction freezes these finite local requirements in a progression before applying Sun's theorem. That theorem allows avoidance at every fixed odd prime of this progression, including ℓ; avoidance at seven follows from its prescribed Gaussian congruence. The reduction-order Chinese remainder conditions remain compatible with the fixed progression. They also preserve the chosen real neighborhood. Thus all three comparison poles and the scalar pole are realized together, exactly with the assigned orders $10, 39$ and scalar order 20. Larger fixed local depths require no new rational coordinate. This completes the realization argument. □

5.4 The exact polynomial and its bounds

The source has 28 retained rational coordinates: the input, 18 point-abscissa coordinates, one slope, six comparison-pole coordinates, and two scalar-pole coordinates. Its remaining finite choices are nine monic quartic point roots, four monic quadratic pole roots, and four bits choosing rational conic charts. The finite-algebra construction eliminates the thirteen roots with one further rational coordinate. Every zero and nonzero quadratic-root case is included. The maximum finite-algebra rank is $4^9 2^4 = 2^{22}$.

Theorem 5.4. *Under the four cited arithmetic inputs there is an explicitly specified polynomial $F \in \mathbb{Z}[X, Y_1, \dots, Y_{28}]$ such that*

$$X \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists (Y_1, \dots, Y_{28}) \in \mathbb{Q}^{28} \quad F(X, \boldsymbol{Y}) = 0.$$

It satisfies

$$\deg F \le 2^{174}, \qquad \max |\operatorname{coeff}(F)| \le 2^{\,2^{177}}.$$

The specification uses determinant, trace, resultant, coefficient extraction and explicitly bounded finite products of integer polynomial expressions. It has no finite product of elliptic-index exclusions.

Proof. The preceding local and height arguments prove soundness; the preceding realization lemma and the simultaneous comparison construction prove completeness. The guarded finite-algebra theorem applies to the stated triangular monic relations and finite conic choices, leaving precisely $28+1=29$ external coordinates. Its determinant and trace formulas have integer coefficients and introduce no quantified matrix entries.

The coefficient directed acyclic graph, all monic relations, every denominator guard, and the finite-product formula are supplied in `polynomial-specification.json`. The independent checker reconstructs every affine conversion from the rational source, verifies all repeated-root branches, and recomputes the degree and coefficient majorants by integer arithmetic. For auditability, its intermediate bounds are

$$\begin{aligned} \text{matrix-entry degree} &\le 4790038912, \\ \log_2(\text{weighted operator norm}) &\le 49766687378, \\ \text{number of final factors} &= 12292021203320302469120. \end{aligned}$$

The resulting exact degree majorant has binary length 174, and the exact majorant for the logarithm to base two of the coefficient norm has binary length 177. The weighted companion-norm lemma and the elementary product, determinant and resultant estimates in the finite-algebra theorem therefore give the stated rounded bounds. This computation certifies the specified algebraic construction and bounds; the four arithmetic theorems remain its explicitly cited inputs. □

5.5 Simultaneous completeness

Fix $X \in \mathbb{Z}$. The input-map lemma gives an odd integer $\beta > 2401$. Lemma 5.3 proves that the scalar contact R_β is positive and less than one, has square reduced denominator, and is square at $2, 3, 17$ with depths at least $4, 4, 2$. Its finite support includes the fixed prime ℓ.

Choose a real compact neighborhood for H_0 near, but away from, the identity in both embeddings, with $f(H_0)$ small and $z(H_0)$ bounded away from zero. The positive orbit of every suitable finite-index subgroup of $\mathbb{Z}P$ is dense in the identity component of its real closure. Thus one may choose a positive h_* divisible by 64, and deep in every formal group over the finite support of R_β, whose point lies in this neighborhood. Choose a positive integer s divisible by six, still deeper at these places, with the fixed prime depth required in Lemma 5.2, and with $[s]P$ sufficiently close to the real identity. Set $a = s + 1$, so a is odd and $a \equiv 1 \pmod 3$. Take $S = [a]P$, and use $C_S = [s]P$ in the scalar certificate. The normalized leaves of its norm tree are then small in the real absolute value and sufficiently deep locally, uniformly for H_0 in a smaller neighborhood. Its root is a local unit and $|W - 1| < 1/2$.

Odd Lucas labels are available explicitly. Let $U_0 = 0$, $U_1 = 1$, $U_{j+1} = aU_j - U_{j-1}$, and put

$$j = 2(a - 2), \qquad b = U_j/(a - 2), \qquad t = U_{j+1}.$$

Modulo $a - 2$ one has $U_j \equiv j$, so b is integral. Modulo two the sequence has period $0, 1, 1$. Since $a \equiv 1 \pmod 3$, $j \equiv 1 \pmod 3$, so b, t are odd. They are positive and greater than one. The identity $U_{j+1}^2 - aU_jU_{j+1} + U_j^2 = 1$ gives $\Delta = 0$. The four root labels β, a, t, b are therefore all odd and at least three, and Lemma 4.4 covers their points.

Temporarily set $H_0 = [h]P$ and $U_i = [hn_i]P$. Local continuity provides a progression $h \equiv h_* \pmod L$, with $64 \mid L$, preserving the scalar units and fixed lower bounds for all comparison tests at $2, 3, 17$. Include the order of the finite component group of the real closure in L. Remove the finitely many comparison pole loci from the real neighborhood and pass to a compact subset; the tests are then uniformly bounded there. These choices precede the prime-pattern application.

Choose D_0 with $D_0^2\theta/A_r \in \mathbb{Z}$ for every contact and scale, put $v = D_0^2w^2$, and choose $\lambda = u/v$. The contact numerators $u - (\theta/A_r)v$ are a finite set of distinct prime forms after repetitions are removed. Apply the simultaneous prime-pattern hypothesis beyond all fixed bad primes, with enough divisibility of w and a sufficiently large lower bound on u/w^2. The curve has full rational two-torsion at every selected residue field. Because P is a double, $d_\pm(p)/4$ annihilates its reduction. Define

$$U = \operatorname{lcm}_p(d_-(p)/4), \qquad V = \operatorname{lcm}_p(d_+(p)/4).$$

The order and avoidance hypotheses give $\gcd(U, V) = 1$, $\gcd(U, L) = 16$, and $\gcd(V, L) = 1$. Thus

$$h \equiv 0 \pmod U, \qquad h \equiv 1 \pmod V, \qquad h \equiv h_* \pmod L$$

are compatible. Their progression still has the required real closure, so H_0 can lie in the chosen compact set. At one place over each contact prime the points H_0, U_i are formal; at the conjugate place V_0, V_i are formal. Both exact logarithmic ratios are n_i. All sixteen tests therefore have their assigned orders, including the direct Lucas polynomial since $\Delta = 0$.

For completeness of each comparison pole, note the exact identity

$$R_r = \frac{\prod_{\theta\in\Theta}(u-(\theta/A_r)v)}{v^5(A_r\lambda-1)^6}.$$

The denominator is a rational square. At the five contact primes $v_p(R_r)=1$. At either place over such a prime each inverse norm factor has residue zero or one, and at the successful place every residue is one. Its reciprocal trace is thus a unit with residue one or two. Consequently b_r has positive odd valuation only at these primes and has square reduced denominator. Their prescribed congruences make b_r square at $2,3,17$.

Uniformly on the compact real set, $R_r \sim A_r^{-6}\lambda^{-1}$, $W_r = 1+O(\lambda^{-3})$, and $b_r \sim (4A_r^6\lambda)^{-1}$. Hence $0<b_r<17$ for a sufficiently large prescribed lower bound on λ. At $p\in\{2,3,17\}$ put $\nu = v_p(v)$ and $\eta = v_p(A_r)$. Sufficient divisibility of w gives $v_p(R_r) = \nu - 6\eta$. The nonconstant terms of the norm factors have valuations at least $3\nu+(10m_j-8)\eta-2c_p$, where c_p is the previously fixed lower-bound constant. Their inverse trace therefore has valuation $v_p(2)$, and $v_p(b_r) = \nu-6\eta-2v_p(2)$. These valuations can exceed every required preparation threshold. The pole hypothesis supplies the three comparison certificates.

The scalar pole input retains square denominator and the specified local squareclasses because W is a unit at every prime in its fixed support. Also $0<R_\beta W^2<4<17$. The same pole hypothesis supplies its certificate with preparation zero. All guards are met, and the conic parametrization covers all these witnesses. This proves completeness for every integer input.

5.6 The finite polynomial and its bounds

The thirteen formal roots consist of nine point quartics and four pole quadratics. A rational equation $q^2 = N/D$ with $D\neq 0$ is made monic by $Z = Dq$, so $Z^2 - ND = 0$. Its zero-radicand branch is $Z=0$ with $ND=0$; its other branch has $Z^2 = ND$ and $Z\neq 0$. Together with the point derivative guards this makes every admitted geometric point simple. The rank is at most

$$4^9\,2^4 = 2^{22}.$$

Theorem 3.1 eliminates all thirteen roots using one new rational variable U. For completeness, its literal recipe is as follows. On each quadratic support branch, put $m=13$ and let $n = 4^9 2^k$, where k is the number of nonzero quadratic branches. Let G be the product of the source guards and E_i the closed zero-radicand conditions. For every $0\le \ell \le mn(n+1)/2$ and $1\le d\le n$, form

$$\lambda_\ell = 1+\sum_{i=1}^{m}\ell^i z_i,\quad H(V)=\det(I-VM_{G\lambda_\ell}) = \sum a_jV^j,\quad R_i(V) = \operatorname{Tr}(M_{E_i}\operatorname{adj}(I-VM_{G\lambda_\ell})).$$

Put $H_d = \sum_{j=0}^{d} a_jV^j$, $g = a_d\operatorname{Res}(H_d,H_d')$, $L = a_d^2 g$, and take the factor

$$\left(1+\sum_{j=1}^{d} a_{d-j}a_d^{2j-1}g^jU^j\right)^2 + \sum_{j>d}a_j^2 + \sum_i\Big((LU)^{n-1}R_i(1/(LU))\Big)^2.$$

Coefficient reversal makes the last expressions polynomials at $LU=0$. Multiply these factors over the displayed ranges, the 2^4 quadratic masks, and all 2^4 original finite choices. This is a completely specified integer polynomial in the retained coordinates and U, with no formal root left as a free variable.

5.7 Weighted companion norms

The coefficient estimate can be sharpened without changing the polynomial. The useful norm on a monic finite algebra assigns different positive scales to its basis monomials. We describe this choice and its compatibility with all determinant and trace operations.

Let $B = \mathbb{R}[X_1, \ldots, X_r]$, equipped with the coefficient norm $\|f\|_1 = \sum_\alpha |[X^\alpha]f|$. This norm is submultiplicative. Consider the free iterated monic algebra

$$A_0 = B, \qquad A_i = A_{i-1}[z_i] / \left(z_i^{d_i} + \sum_{j<d_i} c_{ij} z_i^j \right).$$

Fix the usual basis of monomials $z_1^{e_1} \cdots z_i^{e_i}$, $0 \le e_j < d_j$. Coefficients c_{ij} belong to A_{i-1}. Integer coefficients are a special case of this real coefficient ring; no assertion about zeros over a larger field is being made.

Lemma 5.5. *Suppose the coefficient multiplication operators in A_{i-1} have bounds $\|M_{c_{ij}}\| \le 2^{b_{ij}}$, with $b_{ij} \ge 0$. Let q_i be the number of nonzero coefficients, or any upper bound for that number. Take $q_i \ge 1$, and put*

$$\gamma_i = \max\left(\{0\} \cup \left\{ \left\lceil \frac{b_{ij}}{d_i - j} \right\rceil : c_{ij} \ne 0 \right\} \right) + \lceil \log_2 q_i \rceil.$$

On A_i use the norm

$$\left\| \sum_{j=0}^{d_i-1} a_j z_i^j \right\| = \sum_{j=0}^{d_i-1} 2^{j\gamma_i} \|a_j\|.$$

Then $\|M_{z_i}\| \le 2^{\gamma_i}$. Multiplication by an element of A_{i-1} retains its preceding operator-norm bound.

Proof. For a basis block z_i^j with $j < d_i - 1$, multiplication by z_i shifts to the next block and multiplies its norm weight by 2^{γ_i}. On the last block, the monic relation gives a column of norm at most

$$\sum_{j<d_i} \|M_{c_{ij}}\| 2^{(j-d_i+1)\gamma_i} = 2^{\gamma_i} \sum_{j<d_i} \|M_{c_{ij}}\| 2^{-(d_i-j)\gamma_i}.$$

Write $u = \lceil \log_2 q_i \rceil$. Every nonzero summand in the last sum is at most $2^{-(d_i-j)u} \le 2^{-u}$, and there are at most q_i summands. The sum is therefore at most one. These are all block columns of the companion matrix, proving the desired bound. An earlier multiplication operator acts independently and identically on every block. Its block weights cancel, so its preceding norm bound is unchanged. □

Starting with $\|f\|_1$ on B, the lemma defines a norm throughout the monic tower. The triangle and product inequalities bound any coefficient expression recursively. In particular, every integer constant a has norm $|a|$, every base coordinate has norm one, and an arithmetic expression with recorded logarithmic bound b has operator norm at most 2^b.

For completeness, these weighted operator bounds can be used in the coefficient estimates for the characteristic and trace polynomials without introducing any extra factor for the size of the basis scales. Let D be the constant diagonal matrix whose diagonal entry for a basis monomial is

$$\prod_i 2^{e_i\gamma_i}.$$

The weighted maximum column norm of M is exactly the ordinary maximum column sum of the coefficient norms of DMD^{-1}. Simultaneous conjugation leaves

$$\det(I - VM), \qquad \operatorname{Tr}(M_E \operatorname{adj}(I - VM))$$

unchanged. The adjugate identity follows first for invertible matrices from $\operatorname{adj}(A) = \det(A)A^{-1}$ and then for all matrices by polynomial identity. Thus every determinant, trace and resultant estimate already stated may be applied to the conjugated matrices. The coefficients of the resulting scalar polynomials are exactly the original coefficients. In particular, no additional basis-height term is needed. The weighted degree calculation is separate and is unchanged.

If a quadratic generator is set to zero in one of the finite branches, its norm is zero. The recursive estimates for all later expressions remain valid after that substitution. Thus a single table also bounds all of the zero and nonzero quadratic cases.

For the construction with 29 variables, this gives the generator logarithmic norm bounds

$$\underbrace{5, \ldots, 5}_{9 \text{ entries}}, \quad 89263067, \quad 1539436917, \quad 2988679217, \quad 19861198427.$$

The weighted degree table is unchanged. The multiplication operator used in the determinant has logarithmic norm bound 49766687378. The exact final bound for the logarithm to base two of the coefficient norm is

$$180551808614174879228877281249630444945389422609694720 < 2^{177}.$$

Consequently the same polynomial satisfies

$$\max |\operatorname{coeff}(F)| \leq 2^{2^{177}}.$$

The independent checker recomputes these bounds from the full integer coefficient graph. A separate equality certificate verifies that removing the bound annotations leaves precisely the same mathematical specification. The degree bound remains 2^{174}.

5.8 Explicit degree and coefficient bounds for the specification

The following bounds concern the expanded polynomial defined by the indexed determinant formula. They do not require expanding its matrices or its final finite product. For a polynomial f, let $\|f\|_1$ denote the sum of the absolute values of its coefficients. Then $\|fg\|_1 \leq \|f\|_1\|g\|_1$ and $\|f + g\|_1 \leq \|f\|_1 + \|g\|_1$.

Give every base coordinate weight one, and assign a positive integer weight w_i to each algebra generator z_i. Choose the weights so that every nonleading term of its monic relation has weight at most $e_i w_i$. Reduction by these relations never increases weighted degree. Writing $W = \sum_i (e_i - 1)w_i$, the entries of a multiplication matrix M_f therefore have base degree at most $\operatorname{wdeg}(f) + W$: the input basis monomial has weight at most W, and the output basis monomial has nonnegative weight.

Use the weighted multiplication-operator norms of Lemma 5.5. Their triangle and product inequalities give the coefficient bounds $\|M_{z_i}\| \leq 2^{b_i}$ by recursion. The weights used for polynomial degrees are separate: the point-quartic generators have degree weight two, while each quadratic generator may be assigned half its constant coefficient's weighted degree, rounded upward. Its zero branch satisfies the same bounds. A single table therefore covers all 2^4 quadratic cases. The constant diagonal conjugation proved in the preceding subsection justifies using these smaller operator norms in every scalar determinant and trace estimate below.

Put $n = 2^{22}$ and $m = 13$, and $T = mn(n + 1)/2$ for the largest separating parameter. Suppose the recorded bounds give

$$\operatorname{wdeg}(G) \leq w_G, \quad \|M_G\| \leq 2^{b_G}, \qquad \operatorname{wdeg}(E_i) \leq w_E, \quad \|M_{E_i}\| \leq 2^{b_E}.$$

Set

$$\begin{aligned} b_\lambda &= \lceil \log_2(m + 1) \rceil + \max_i \{ i \lceil \log_2 T \rceil + b_i \}, \\ \alpha &= w_G + \max_i w_i + W, \qquad \beta = b_G + b_\lambda, \\ \alpha_E &= w_E + W. \end{aligned}$$

These bound the entry degrees and operator norm of $M_{G\lambda_\ell}$, uniformly for all prescribed ℓ and all source branches.

The coefficients of $H(V) = \det(I - VM)$ have base degree at most $A_H = n\alpha$, and its full coefficient norm is at most 2^{B_H} with $B_H = n(\beta + 1)$. Indeed the coefficient norm of a determinant is bounded by the product of its column sums. This bound already includes the sum over determinant permutations; no additional factorial is needed. By multilinearity in columns,

$$R_i(V) = \frac{\partial}{\partial t} \det(I - VM + tM_{E_i})\Big|_{t=0}$$

has base coefficient degree at most $A_R = (n-1)\alpha + \alpha_E$ and coefficient norm at most 2^{B_R}, where

$$B_R = \lceil \log_2 n \rceil + b_E + (n-1)(\beta + 1).$$

This accounts for the n possible columns containing M_{E_i}.

For each degree case $d \le n$, the Sylvester matrix for $\mathrm{Res}(H_d, H_d')$ has $2d - 1$ rows. Every row has coefficient norm at most $d\|H\|_1$. Hence uniform bounds for the reciprocal scale $L = a_d^3 \,\mathrm{Res}(H_d, H_d')$ are

$$\begin{aligned} A_L &= (2n+2)A_H, \\ B_L &= (2n+2)B_H + (2n-1)\lceil \log_2 n \rceil. \end{aligned}$$

The reciprocal residual has degree at most $n(A_L+1)$ and coefficient norm at most $2^{nB_L + \lceil \log_2(n+1) \rceil}$. The reversed trace residuals have degree at most $A_R + (n-1)(A_L+1)$ and coefficient norm at most $2^{B_R + (n-1)B_L}$. Together with the high-coefficient conditions $a_k = 0$, these give a bound for the sum of squares by doubling the largest degree, doubling the largest logarithmic norm bound, and adding $\lceil \log_2(n+q+1) \rceil$ to the latter. For the reciprocal estimate, no division of coefficient norms is used: its jth coefficient is the polynomial $a_{d-j} a_d^{3j-1} \,\mathrm{Res}(H_d, H_d')^j$. Its degree is at most jA_L and its logarithmic norm bound is at most $j(3B_H + B_{\mathrm{Res}}) = jB_L$.

The number of final factors is also explicit. Put $n_0 = 4^9$ and $q = 4$. For a mask with k nonzero quadratic branches, the rank is $n_k = n_0 2^k$. There are $\binom{q}{k}$ such masks, each with $n_k(1 + mn_k(n_k+1)/2)$ separating-parameter and degree choices. Including the 2^4 original finite parameter choices gives exactly

$$B_{\mathrm{tot}} = 2^4 \left[n_0 3^q + \frac{m}{2}(n_0^3 9^q + n_0^2 5^q) \right]$$

factors. Its exact value is

$$B_{\mathrm{tot}} = 12292021203320302469120.$$

Multiplying the uniform branch-degree and logarithmic coefficient-norm bounds by this integer bounds the final product. All the required weights and norms are computed by finite recursions on the supplied coefficient graph. The exact integer majorants give

$$\deg F \le 2^{174}, \qquad H(F) \le 2^{2^{177}}.$$

These estimates require no expanded matrix or monomial table. The independent checker evaluates the recursions and compares every integer bound in the saved specification.

6 The input map and its local squareclasses

Put

$$\beta(X) = 216X^4 + 450459$$

and retain the five-contact polynomial

$$r(T) = T(T-1)(T-2401)(32T+343)(2T-1).$$

Define

$$r_X = r(\beta(X)).$$

Lemma 6.1. *For $X \in \mathbb{Q}$ one has*

$$\beta(X) \in \mathbb{Z} \quad \Longleftrightarrow \quad X \in \mathbb{Z}.$$

For every real X, $\beta(X) \geq 450459 > 2401$, so $r_X > 0$. If $p > 76176$ and $v_p(\beta(X)) < 0$, then

$$v_p(\beta(X)) = 4v_p(X), \qquad v_p(r_X) = 5v_p(\beta(X)).$$

In particular, both valuations are even.

Proof. If X is an integer, so is $\beta(X)$. Conversely, write $X = a/b$ in lowest terms with $b > 0$. Integrality of $\beta(X)$ gives $b^4 \mid 216 = 2^3 \cdot 3^3$. Neither prime exponent admits a nontrivial fourth power, so $b = 1$.

The real inequalities follow from the fourth power and the five positive factors of $r(T)$ when $T > 2401$. At a prime in the stated range, a negative valuation of β forces $v_p(X) < 0$. The unique dominant terms in β and $r(\beta)$ are respectively $216X^4$ and $64\beta^5$. These identities prove the claimed valuation formulas. □

Lemma 6.2. *If $X \in \mathbb{Z}$, then r_X is a nonzero square in $\mathbb{Q}_2$, $\mathbb{Q}_3$, and $\mathbb{Q}_{17}$. More precisely,*

$$v_2(r_X) = 2, \quad r_X/4 \equiv 1 \pmod 8, \qquad v_3(r_X) = 2, \quad r_X/9 \equiv 1 \pmod 3,$$

and at seventeen either $v_{17}(r_X) = 0$ with square-unit residue, or

$$v_{17}(r_X) = 2, \qquad r_X/17^2 \equiv 9 \pmod{17}.$$

Proof. Since 216 is divisible by eight and by twenty-seven,

$$\beta \equiv 3 \pmod 8, \qquad \beta \equiv 18 \pmod{27}.$$

For the two even factors of $r(\beta)$ one has

$$v_2(\beta - 1) = v_2(\beta - 2401) = 1, \qquad \frac{\beta - 1}{2} - \frac{\beta - 2401}{2} = 1200 \equiv 0 \pmod 8.$$

Their odd parts therefore have square product congruent to one modulo eight. The remaining odd factors have product $3 \cdot 7 \cdot 5 \equiv 1 \pmod 8$. This proves the assertions at two. At three, only the factor β is divisible by three; it has valuation two. Reduction after division by nine gives

$$\frac{r(\beta)}{9} \equiv 2 \cdot 2 \cdot 2 \cdot 1 \cdot 2 \equiv 1 \pmod 3.$$

At seventeen the fourth powers are exactly $0, 1, 4, 13, 16$. Since $216 \equiv 12$ and $450459 \equiv 10 \pmod{17}$, all possibilities are given by the table

X^4	0	1	4	13	16
β	10	5	7	13	15
$r(\beta)$	0	15	2	13	2.

The nonzero residues are squares: $13 = 8^2$, $15 = 7^2$, and $2 = 6^2$ modulo seventeen.

In the remaining case $17 \mid X$. The exact identity

$$32 \cdot 450459 + 343 = 17^2 \cdot 49879, \qquad 49879 \equiv 1 \pmod{17},$$

and $v_{17}(X^4) \geq 4$ show that the factor $32\beta + 343 = 6912X^4 + 17^2 \cdot 49879$ has valuation exactly two and unit residue one. The other four factors are units, with product $10 \cdot 9 \cdot 6 \cdot 2 \equiv 9 \pmod{17}$. This proves the last assertion. The usual local square criteria at two and at odd primes now prove all three square statements. □

7 Number-field scope

All displayed algebraic identities remain valid in characteristic zero. Positivity of the contact and norm root also holds in a number field with a real embedding, after applying that embedding. The local squareclass table, however, concerns rational integer inputs and the three specified completions of $\mathbb{Q}$. It does not establish the corresponding preparation assertions for every algebraic integer in a larger field.

Even the integrality equivalence of Lemma 6.1 does not automatically become an equivalence for rings of integers. Let x_0 be a real fourth root of $1/108$ and $L = \mathbb{Q}(x_0)$. Then

$$\beta(x_0) = 450461 \in \mathbb{Z}.$$

But x_0 is not an algebraic integer, since $x_0^4 = 1/108$. Thus this elementary input map alone does not transfer a definition to $\mathcal{O}_L$. No assertion that the complete rational defining polynomial defines every $\mathcal{O}_L$ is made or follows from this normalization.

8 Exact specification and reproducibility

The directory `computation/few_variables/forced_prime29/` contains the rational source, its producer, the integer polynomial specification, and independent checkers. The reproduction script rebuilds the source and specification byte for byte, checks all monic relations and denominator guards, and recomputes the degree and coefficient estimates. Separate integer checks certify the chosen prime, all five fourth-power exclusions, and the strict index margin.

The determinant and resultant matrices and the final finite product are specified exactly; their full expansions are not stored. The global arithmetic inputs remain the four stated premises of the projection theorem. No computational certificate is used as a replacement for one of those theorems.

References

[1] E. Glazer and contributors, *Diophantine definition of the integers in the rationals*, source repository, coefficient baseline `14e8f601fe0d9173d4a5f084aedc3e07066a7688`, 7 November 2026. The odd-product identity, direct targets and asymmetric-depth construction are in `docs/quartic-523-540/addition.tex`; the repository attributes the odd-product contribution to G. Conant, developed with ChatGPT. https://github.com/elliotglazer/diophantine-z-q/tree/14e8f601fe0d9173d4a5f084aedc3e07066a7688.

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